The unlucky seed was an untrained seed. The band was not a budget artifact

N=48, degree 5, 4 colours, leak-free | one knob: 50 -> 100 epochs, 375k -> 750k steps, fixed 240k pool | 84 cells

WHY A BUDGET ARM EXISTS AT ALL

Across stages A-F, 12 of 24 multi-seed rungs never got below finding 44's 1.64 train-NLL plateau, and each was written off as 'did not train in 375k steps' -- deliberately NOT as evidence that its architecture lacks the capacity. That caveat is honest and it is unfalsifiable until someone spends the steps. Stage G spends them on the cheapest rung in the grid (G1, attention-only d128, 480,058 trainable) and on the rung that was the standing minimum when the job was written (G2, attention-only d256, 1,746,490).

WHAT 750k STEPS BOUGHT: ONE WHOLE SEED, AND THE SPREAD

G2's seed 1 was a NON-SOLVER at 50 epochs -- censored, best solve 0.465 at any hint density, train NLL 1.688, sitting on the plateau. At 100 epochs it is rho_50 0.761, best solve 0.963: indistinguishable from its two siblings. The rung goes 2 of 3 crossing seeds to 3 of 3, and its seed spread in mean_solve COLLAPSES from 0.2671 to 0.0041. The spread at this rung was a budget artifact.

WHAT IT DID NOT BUY: THE METRIC OF RECORD

Median rho_50 0.763 -> 0.762, a move of 0.001 against the frozen seed-spread bar 0.1484, and mean_solve 0.3039 -> 0.3070. The median was already set by a seed that had trained, so rescuing the third one moves the rung nowhere: THE 0.76 BAND IS NOT A BUDGET ARTIFACT. G1 is unchanged on every count -- 0 of 3 escapes against its twin's 0, rho_50 CENSORED -- and by the registered asymmetry that is still 'did not train', not 'cannot'.

rho_50 = realized hint density at which solve rate crosses 0.5 on a 14-point x 400-instance grid, LOWER IS BETTER, reported CENSORED when no bin crosses. mean_solve = unweighted mean over the 14 bins, which never censors. s* = 0.1484 / 0.2918 was frozen before any of these cells ran and is not recomputed. Commit dd9c400.

What the model actually sees

Every token is one categorical id into a 58-symbol vocabulary (N=48, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING
	BEGIN_ROUND SOLVED UNRESOLVED
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3
	the q colours (token name COLOR_n, prose 'colour')
10-57	N0 N1 ... N47
	one id per node, no structure in the id

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*

SOLVED

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 401 TOKENS

HINTS	BEGIN_HINTS N0 COLOR_0 N1 COLOR_2 N2 COLOR_3 N10 COLOR_1 N15 COLOR_1 N17 COLOR_3 ... (37 tokens)
GRAPH	BEGIN_GRAPH N0 N16 N8 N38 N34 N17 N1 N13 N27 N34 N28 N10 ... (289 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N9 COLOR_3 N14 COLOR_2 N16 COLOR_2 N32 COLOR_0 N34 COLOR_1 N44 ... (75 tokens)

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 13 rounds, 30 deductions,
60 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000018, FALL-colouring ensemble (every node sees all q-1 other colours, so no node is permanently undeducible).

One real reasoning trace

18 hints of 48 nodes, then 13 rounds of simultaneous deduction. Hints: N0=COLOR_0 N1=COLOR_2 N2=COLOR_3 N10=COLOR_1 N15=COLOR_1 N17=COLOR_3 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND      N9 COLOR_3 N14 COLOR_2 N16 COLOR_2 N32 COLOR_0 ...
  6 deduced
BEGIN_ROUND      N42 COLOR_3 N45 COLOR_1
  2 deduced
BEGIN_ROUND      N13 COLOR_0 N20 COLOR_0 N37 COLOR_2
  3 deduced
BEGIN_ROUND      N21 COLOR_2 N24 COLOR_3 N31 COLOR_2
  3 deduced
... 9 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N9, neighbours N25 N10 N29 N31 N18

their colours known before this round:

```
[0, 1, 2]  (3 = q-1 of the 4)

(3 known neighbours, 3 DISTINCT colours)

still unknown: N25 N31
```

only colour left: 3

```
so the trace emits  N9 COLOR_3
```

The model is never shown this reasoning -- only the node/colour pair.

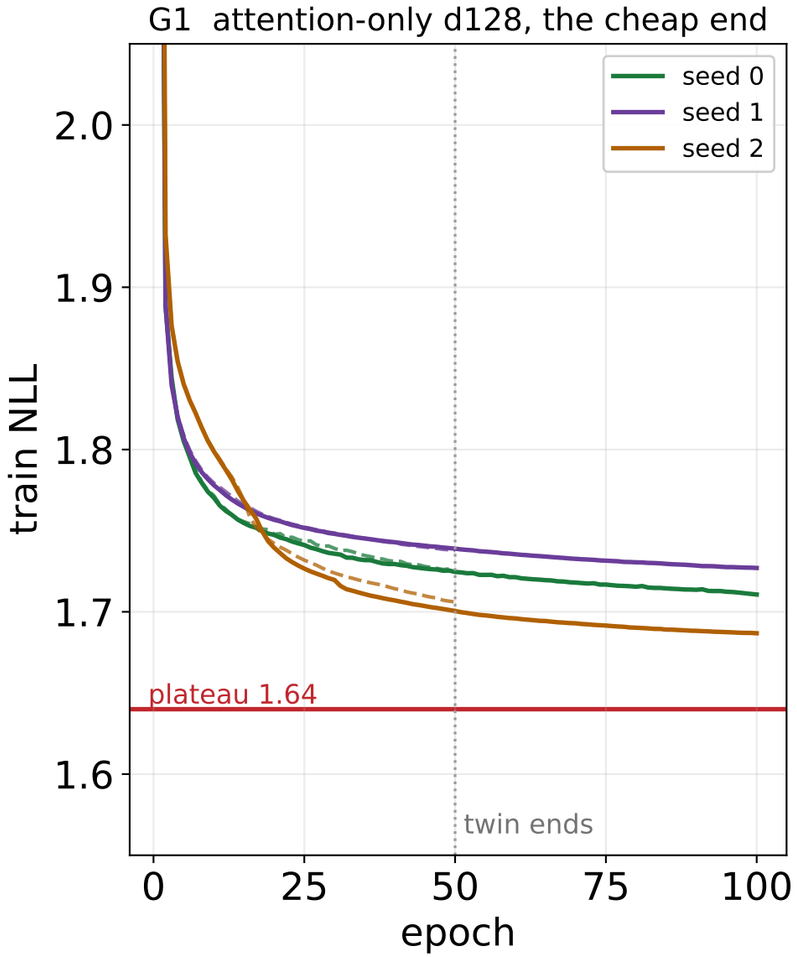
A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

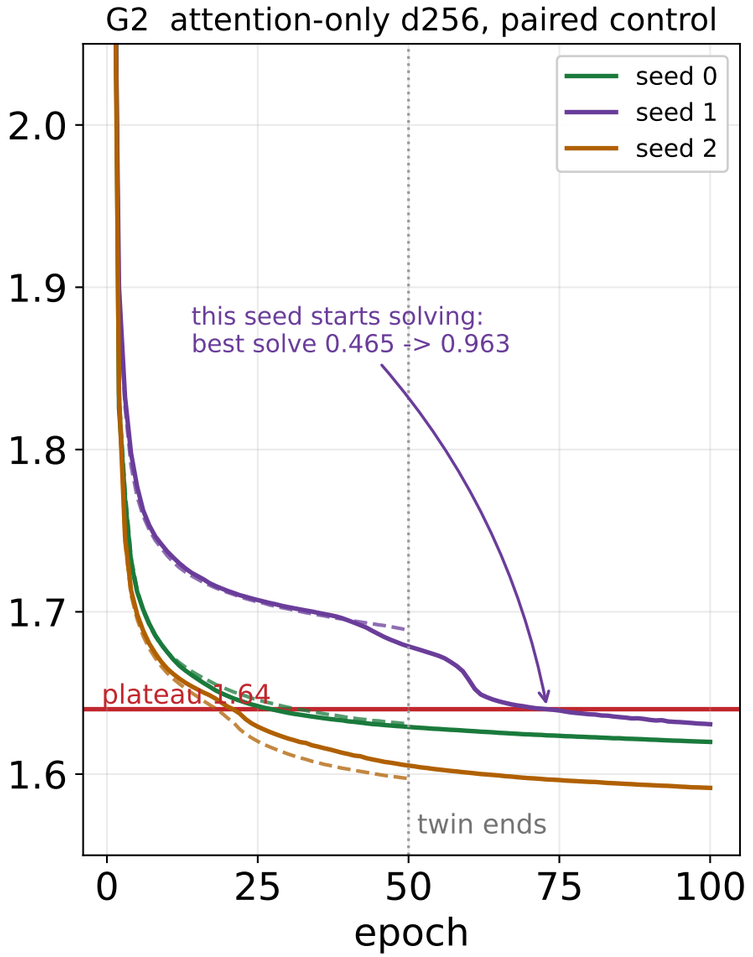
N=48, degree 5, 4 colours, --shuffle-adj --fall: the task every cell in this deck is trained and scored on. Stage G changes nothing about it -- the same 240k instances are shown twice as many times, which is the only difference between a 50-epoch cell and its 100-epoch twin.

T=13 rounds, the 90th percentile of 40 draws from this config (range 1-16) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

Twice the steps, same seeds, same pool: where the extra 375k went

Dashed = the 50-epoch twin, solid = the 100-epoch run. They are the same optimization until epoch 50, and that is checked, not assumed





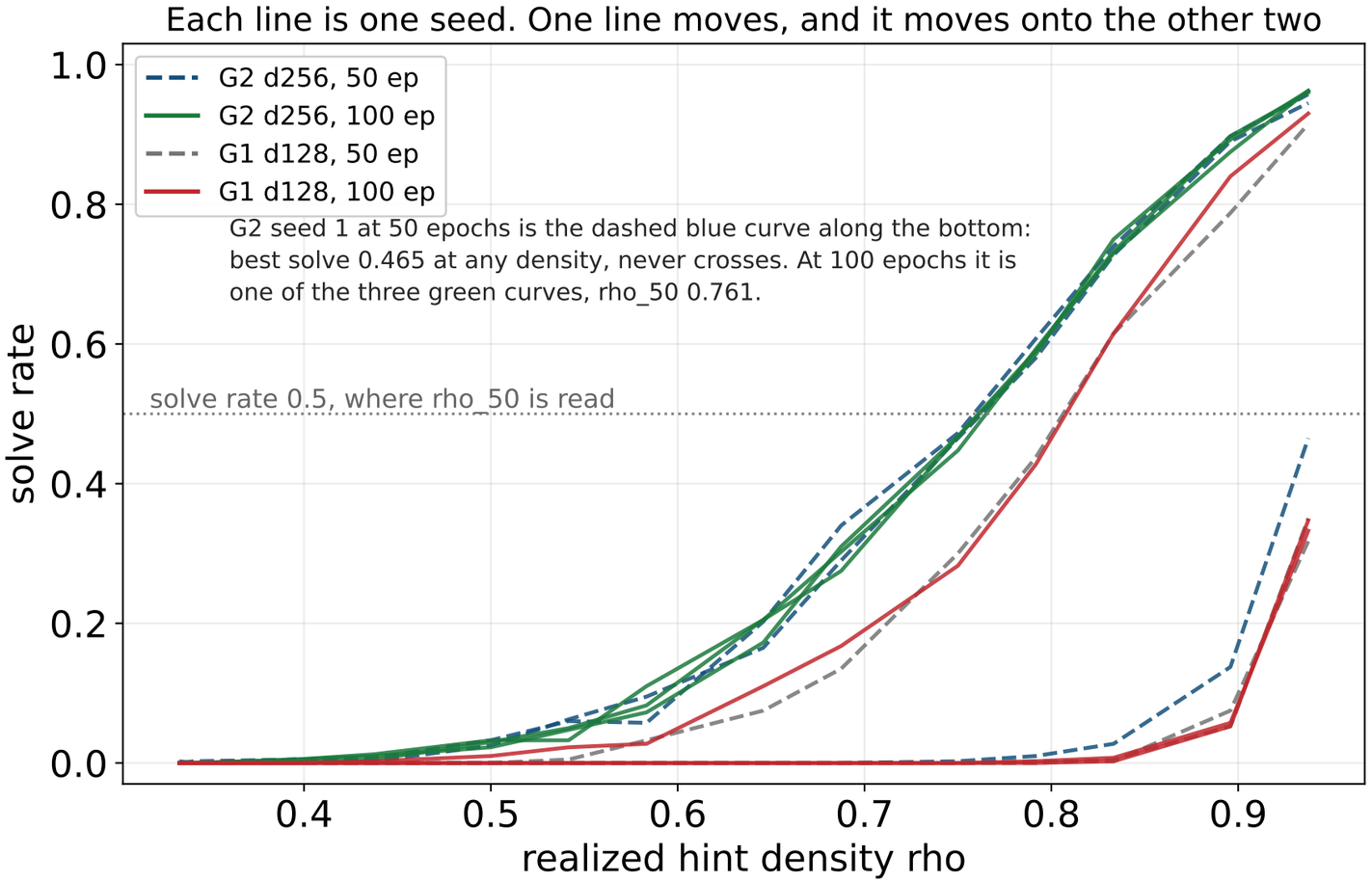
EPOCH-50 REPRODUCTION GATE
tolerance 0.01 nats, registered before the job

	seed	epoch-50	twin final	delta	
G1					
	0	1.725	1.725	-0.000	ok
	1	1.739	1.738	+0.001	ok
	2	1.701	1.706	-0.006	ok
G2					
	0	1.629	1.631	-0.002	ok
	1	1.679	1.688	-0.010	ok
	2	1.605	1.597	+0.008	ok

The y axis is clipped at 2.05; the six curves start between 2.20 and 2.60 at epoch 1. With a constant learning rate, no warmup, a fixed pool and the same seed, the first 50 epochs of a 100-epoch run ARE the 50-epoch run. If a cell missed the gate the pairing would be void and the cell reported inconclusive rather than explained -- nothing here missed it. Note what the escape looks like: G2 seed 1 drifts across the line rather than turning a corner, which is what escape_timing.py measured on all 14 late escapes in the grid. Commit dd9c400.

The rescued seed lands on top of its siblings, and the median stays put

Transition curves on the identical 14-point x 400-instance grid, three seeds per rung, --shuffle-adj throughout



	50 ep	100 ep
G1 attention-only d128		
rho_50	CENSORED	CENSORED
crossing seeds	1/3	1/3
mean_solve	0.0291	0.0287
seed spread	0.2075	0.2168
token accuracy	0.4614	0.4632
plateau escapes	0/3	0/3
G2 attention-only d256		
rho_50	0.763	0.762
crossing seeds	2/3	3/3
mean_solve	0.3039	0.3070
seed spread	0.2671	0.0041
token accuracy	0.4764	0.4830
plateau escapes	2/3	3/3

trainable 1,746,490, unchanged
and asserted against the twin

REGISTERED THRESHOLD, from the scorer:
G2 below rho_50 0.615 would have
made the 0.76 band a budget artifact.
It is 0.762.

The comparison that is legitimate here is each rung against its OWN twin: same architecture, same seeds, same pool, one knob. G1's rho_50 against the frozen 0.76 band would be CROSS-BUDGET and the scorer refuses to print it. The effect of 375k extra gradient steps on the standing minimum's rung is a 0.001 shift in median rho_50 (0.01 s*) and +0.0066 in token accuracy -- but a 65x collapse in the seed spread, because the dispersion was one untrained seed and not three different answers. Commit dd9c400.

Two registered predictions disagreed, and the data picked one

Both were in the repository before any stage G cell finished; neither was adjusted afterwards

THE SBATCH HEADER (4a35a2e), WRITTEN AT SUBMISSION

G1 escapes >=2 (trainability) / <=1 (capacity). G2 escapes 3 of 3 -- a guess, made from the fact that two of G2's three seeds were already below the plateau at epoch 50 and the third was still descending.

THE EXTRAPOLATION (dc7c08e), WRITTEN WHILE THE JOB RAN

escape_timing.py's Q3 found that late escapes have no knee -- all 14 in the grid crossed from a train NLL of 1.640-1.652, 13 of 14 within five epochs of the plateau at their own prior rate -- and extrapolating each twin's last-20% slope to epoch 100 predicted G1 0 of 3 and G2 2 of 3. The G2 number openly contradicted the header.

OBSERVED, AND THE HEADER WON ON THE ONE THEY DISAGREED ABOUT

G1 0 of 3, as both predicted. G2 3 of 3, so the header was right and the extrapolation was wrong: seed 1 crossed a gap its own terminal slope put 72 epochs away, and it is the one cell of six to land below its extrapolated bound (G2 seed 1, 1.631 against 1.654). The tails bend down rather than running straight, so Q3's ETAs are UPPER bounds on how far a stalled cell is and its count of 10 cells within another 50 epochs is an undercount. That is one of the two findings here that outlive this rung; the other is on the last page.

The asymmetry is enforced by the scorer, not by the reader: G1 escaping would have proved the budget was binding, G1 not escaping proves nothing, and there is no symmetric reading available. Three seeds per rung, medians compared to medians, against the frozen s* = 0.1484 / 0.2918. Commit dd9c400.

What stage G settles, what it does not, and what is running

The budget question is answered for one rung and left open for the other, on purpose

SETTLED: THE 0.76 BAND IS NOT AN ARTIFACT OF 375k STEPS

The standing minimum's rung was rerun at double the budget with every seed now past the plateau and all 3 of 3 solving, and its median rho_50 moved 0.001 (0.01 s*). Nothing in stages A-F that was scored inside the band needs rereading as a budget effect, and the registered 0.615 threshold that would have forced that rereading was not approached.

SETTLED, AND NOT WHAT THE JOB WAS FOR: WHAT A SEED SPREAD IS

This rung's dispersion was the programme's standing excuse for not reading small gaps -- mean_solve spread 0.2671, wider than most depth steps. At 100 epochs it is 0.0041. The spread was ONE UNTRAINED SEED, not three different answers, and the two seeds that had trained were already within 0.0022 of each other in token accuracy. So a 2-of-3 rung at 50 epochs should be read as 'one seed is not finished', not as 'this architecture is unreliable' -- which is a claim about the frozen s* bar too, since it was pooled at 50 epochs and is therefore conservative, not tight.

NOT SETTLED: WHETHER d128 CAN EVER LEARN THIS

G1 is CENSORED at 0 of 3 escapes after 750k steps. The registered reading is 'did not train', not 'cannot represent it', and the next cell on that question is 200 epochs rather than a conclusion.

UNCHANGED, AND STILL RUNNING

Best on this eval at any size: the complete model at L=12, rho_50 0.385 at 9,638,458 trainable. Standing minimum, by stage H's rule (3 of 3 crossing AND median at or below 0.740): attention-only L=12, 3,331,642 trainable at rho_50 0.605 on 3 of 3 seeds (stage E, 51171824); G3 did not fire, so stage G does not move it. Stage H (51223007) is in flight -- attention-only at L=12, d128 and d192, three seeds each -- asking whether the depth that bought the MLP back survives being made narrow. Its scorer was registered before its first cell landed, as this one was.

Cost, for the record: 6 cells, about 25 GPU-hours, to move one escape count, one seed from censored to solving, and no metric of record. That is the price of being able to write 'the band is not a budget artifact' instead of 'the band is probably not a budget artifact' -- and of learning what the spread it is measured against is made of. Commit dd9c400.